ANALYSTLAB AFRICA | WEEK 6 DATA SCIENCE

Executive Summary

HealthConnect Rwanda | Healthcare data quality, performance analysis, and predictive assessment

5,000	4,443	94.9%	0.916
Appointments	Distinct patients	Facility/department ID mismatch	Monthly receipt-volume correlation

Executive mandate

This Week 6 project integrates 30 healthcare tables into an appointment-level analytical dataset and evaluates patient, clinical, operational, financial, and experience patterns. The aim is to identify decisions supported by the available records and separate them from questions that require better source data.

Key takeaway

The integrated data support descriptive analysis, but several source inconsistencies constrain operational and financial decisions. All 5,000 appointment rows have unique keys and no residual null cells after cleaning; however, zero nulls do not remove unknown clinical histories or validate source values. The most actionable findings are the mapping inconsistency, status/payment mismatch, mixed patient feedback, and weak predictive model performance.

Priority findings

• 4,744 of 5,000 appointments (94.9%) show a disagreement between available facility and department mapping fields.

• 1,131 distinct patients (25.5%) have no recorded chronic-condition history; this is unknown status, not a negative diagnosis.

• Feedback covers 1,756 appointments (35.1%); the mean rating is 2.93/5 and 40.9% of ratings are 1-2.

• The financial time series has 12 monthly observations; receipts and patient volume correlate strongly (rho=0.916), but this is not causal evidence.

Performance and statistical evidence

Question	Result	Decision value
Satisfaction across facilities	Kruskal-Wallis p=0.727; epsilon-squared approximately 0	No evidence of a difference in this sample.
Admission by age group	Chi-square p=0.0448; Holm p=0.179; Cramer V=0.0469	Not significant after family-wise adjustment; small effect.
Severity and stay among admitted patients	Spearman rho=-0.017; p=0.684	No observed monotonic relationship.
Workload proxy and satisfaction	Spearman rho=-0.024; p=0.731	No observed relationship; index is roster-based.
Monthly receipts and patient volume	Spearman rho=0.916; raw p=0.00003; Holm p=0.00014	Strong co-movement across 12 months; no causal claim.

Predictive assessment

The three-month mean baseline outperformed Holt’s damped trend in the final-three-month backtest for visits, gross billed amounts, and recorded receipts. It yields a planning reference of about 412 monthly visits, 62.81 million billed, and 56.84 million receipts. The brief history limits forecast reliability. Admission prediction produced ROC AUC 0.466 and balanced accuracy 0.468; satisfaction classification achieved 0.201 balanced accuracy across five classes. Neither classifier is ready for operational decisions.

Segmentation

Patient K-means selected three clusters (silhouette 0.411). The groups differ in documented chronic history and appointment frequency, but missing-history patterns prevent clinical interpretation as risk tiers. Facility and department clusters show relatively close averages and rely on imperfect location and roster proxies.

Management priorities

Priority	Recommended action	Measure of progress
1. Repair location mapping	Publish and validate DepartmentID-to-FacilityID mappings.	Share of appointments with reconciled identifiers; exception backlog.
2. Reconcile financial records	Align invoice status, payments, balances, and claim lifecycle.	Bill-level reconciliation rate; aging and unexplained balance.
3. Improve data capture	Record chronic-history unknown separately from negative; document currency and timestamps.	Completeness by field and source; unknown rate; lineage coverage.
4. Collect capacity events	Capture staff assignments, shifts, bed occupancy, and equipment availability over time.	Coverage of scheduled and observed capacity by facility/department.
5. Improve patient feedback	Review low-score themes and response rates by site and service.	Response coverage and rating distribution by subgroup.
6. Build modeling evidence	Collect longer history and validated outcomes; use temporal holdout and calibration.	Performance versus baseline, calibration, fairness, and drift.

Six-month execution sequence

Weeks 1-4: confirm data owners, mapping dictionary, billing reconciliation rules, and metric definitions. **Weeks 5-8:** publish a refreshed quality dashboard and collect capacity/feedback metadata. **Weeks 9-16:** repeat descriptive analyses on reconciled data and extend the time series. **Weeks 17-24:** only then evaluate prospective forecasting and risk-support prototypes with clinical and operations review.

Decision statement

HealthConnect Rwanda can use this analysis to prioritize data remediation and focused operational reviews. It should not use the current classifiers for clinical triage, the roster proxies for staffing decisions, or unreconciled amounts as audited revenue. The next milestone is a governed, reconciled dataset with a longer time history and clear ownership for each metric.

Scope and caveats

• The master represents 5,000 appointments and 4,443 distinct patients in the supplied period.

• Source monetary units and currency are unspecified; linked amounts are not independently audited revenue.

• The unlinked Revenue_Summary file was not allocated to departments or appointments.

• InsuranceID availability is not verified active coverage. Chronic history and feedback have incomplete coverage.

• Forecasts and models are exploratory; no external validation or prospective deployment test was performed.

Prepared by Dorian NJAMI | Data Science Intern | AnalystLab Africa | October 2026